

B. Tech. (3rd Sem)

(Common for CSE, CSE with Specialization in Data Science, CSE with specialization in Cloud Technology & Information Security, CSE with specialization in Big Data and Analytics, CSE with specialization in Full Stack Development, CSE with specialization in AI & Machine Learning)

BMAT-003C (Statistical Techniques)

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Continuous evaluation	40
End semester exam	60
Total marks	100
Credits	3.0

Course Objectives:

1. To provide students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science like disease modeling, climate prediction and computer networks etc.
2. To introduce students the concepts, processes and practice of statistical analysis.
3. Understand the essentials of statistical analytics and the corresponding terminologies.

Unit:-1**(9 Hours)**

Foundation of Statistics: Frequency distribution: Frequency distribution and its table and graphical representation, Measures of central tendency: Arithmetic mean, Geometric mean, Harmonic mean, median and mode. Measure of dispersion: Range, deviation (Quartile, mean and standard), Co-efficient of variation.

Unit:-2**(9 Hours)**

Probability Functions and Moment Generating Function: Sample space, Probability set function, Geometric probability. Real random variables - Discrete and continuous, Cumulative distribution function, Probability mass/density functions, Transformations, Mathematical expectation, Moments, Moment generating function, Characteristic function.

Unit:-3**(9 Hours)**

Univariate Discrete and Continuous Distributions: Discrete distributions: Uniform, Bernoulli, Binomial, Negative Binomial, Geometric, and Poisson; Continuous distributions: Uniform, Gamma, Exponential, Beta and normal; Normal approximation to the binomial distribution.

Unit:-4**(9 Hours)**

Correlation and Regression: The Correlation coefficient, Covariance, Calculation of covariance from joint moment generating function, Independent random variables, Linear regression for two variables, Method of least squares, Bivariate normal distribution.

Course Outcome: After completion of this course, students will be able to:

1. Apply some commonly used statistical techniques to other fields.
2. Strengthen themselves both computationally and analytically.
3. Understand and critically discuss the issues surrounding sampling and significance.
4. Solve a range of problems using the techniques covered.
5. Discuss critically the uses and limitations of statistical analysis.

Instructions for paper setter: All Questions are compulsory. The Question paper is divided in to four sections A, B, C and D. Section A is compulsory and comprises of 12 questions of one mark each, 3 from each unit. The questions shall be asked in such a manner that there are no direct answers including one word answer, fill in the blanks or multiple choice questions. Section B comprises of 4 questions of 2 marks each, one from each unit. Section C Comprises of 4 questions of 4 marks each, one from each unit. Section D Comprises of 4 questions of 6 marks each, one from each unit. There is no overall choice, however internal choice may be provided in section C and D, if paper setter so desires.

Text Books:

1. Introduction to Mathematical Statistics (7th ed.) by Hogg, Robert V., McKean, Joseph W., & Craig, Allen T.
2. Mathematical Statistics with Applications (8th ed.) by Miller, Irwin & Miller, Marylees., John E. Freund.

Reference Books:

1. Introduction to Probability Models (11th ed.) by Ross, Sheldon M.